

Chapter 3

Operators and Quantum Dynamics

Lecture 8 developed unitary time evolution and showed that a general state can be expanded in energy eigenstates. This lecture studies the physical consequences of that expansion, introduces inner products and projection coefficients, and solves the infinite square well.

Questions.

- How does a superposition of different energies evolve in time?
- How do orthogonality and inner products determine expansion coefficients?
- Why can interference make a probability density time-dependent?
- How do boundary conditions quantize the energy of an infinite square well?

Lecture roadmap. We begin with energy superpositions and projection coefficients. We then discuss interference and generalized eigenfunctions before solving the infinite square well.

Lecture 9: Energy Superpositions and Bound States

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Review from Lecture 8

For a time-independent Hamiltonian,

$$\hat{H}\phi_n(x) = E_n\phi_n(x). \quad (3.1)$$

If the initial state is

$$\psi(x, 0) = \sum_n c_n \phi_n(x), \quad (3.2)$$

then linearity of the Schrödinger equation gives

$$\boxed{\psi(x, t) = \sum_n c_n \phi_n(x) e^{-iE_n t/\hbar}.} \quad (3.3)$$

Each energy component keeps the same magnitude but acquires its own time-dependent phase.

3.14 One energy versus several energies

3.14.1 One energy eigenstate

A state with one definite energy evolves as

$$\psi_n(x, t) = \phi_n(x) e^{-iE_n t/\hbar}. \quad (3.4)$$

Since the phase has unit modulus,

$$|\psi_n(x, t)|^2 = |\phi_n(x)|^2. \quad (3.5)$$

Thus one energy eigenstate is a stationary state: its probability density is independent of time.

3.14.2 Several energy eigenstates

For a superposition,

$$\psi(x, t) = \sum_n c_n \phi_n(x) e^{-iE_n t/\hbar}. \quad (3.6)$$

Different energies accumulate phase at different rates. The coefficients $|c_n|^2$ remain fixed, but changing relative phases can change interference and therefore change the position probability density.

Questions.

Question 1. Let ψ_1 and ψ_2 be nondegenerate energy eigenstates with $E_1 \neq E_2$. Is $\psi_1 + \psi_2$ also an energy eigenstate?

- A) Yes, always.
- B) No, never.
- C) Possibly; it depends.

Answer and explanation. The answer is **B**. By linearity,

$$\hat{H}(\psi_1 + \psi_2) = E_1 \psi_1 + E_2 \psi_2. \quad (3.7)$$

Because $E_1 \neq E_2$, no single energy can be factored out. The sum is a valid state, but it does not have a definite energy. If the states had the same eigenvalue, their sum would remain an eigenstate.

3.14.3 Operator-order review

The commutator changes sign when its order is reversed:

$$[\hat{p}, \hat{x}] = -[\hat{x}, \hat{p}]. \quad (3.8)$$

Operator addition, however, is commutative.

Questions.

Question 2. Given $[\hat{x}, \hat{p}] = i\hbar\hat{1}$, is $[\hat{p}, \hat{x}] = i\hbar\hat{1}$? Is $i\hat{p} + c\hat{x} = c\hat{x} + i\hat{p}$?

- A) Yes; yes.
- B) Yes; no.
- C) No; yes.
- D) No; no.

Answer and explanation. The answer is **C**:

$$[\hat{p}, \hat{x}] = -[\hat{x}, \hat{p}] = -i\hbar\hat{1}, \quad i\hat{p} + c\hat{x} = c\hat{x} + i\hat{p}. \quad (3.9)$$

The order of multiplication can matter, while the order of addition does not.

3.15 Inner products, orthogonality, and coefficients

3.15.1 Inner product and norm

For two wave functions, define

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f^*(x)g(x) \, dx. \quad (3.10)$$

The norm is

$$\|f\|^2 = \langle f, f \rangle = \int_{-\infty}^{\infty} |f(x)|^2 \, dx. \quad (3.11)$$

Because the first function is complex-conjugated, a constant in the first slot is also conjugated.

Questions.

Question 3. If $f(x)$ and $g(x)$ are wave functions and c is a constant, what is $\langle cf, g \rangle$?

- A) $c\langle f, g \rangle$
- B) $c^*\langle f, g \rangle$
- C) $|c|\langle f, g \rangle$
- D) $c\langle f^*, g \rangle$
- E) None of these.

Answer and explanation. The answer is **B**:

$$\langle cf, g \rangle = \int (cf)^* g \, dx = c^* \int f^* g \, dx = c^* \langle f, g \rangle. \quad (3.12)$$

3.15.2 Orthogonality and the projection formula

For orthonormal energy eigenfunctions,

$$\int_{-\infty}^{\infty} \phi_m^*(x)\phi_n(x) \, dx = \delta_{mn}. \quad (3.13)$$

Here $\delta_{mn} = 1$ when $m = n$ and 0 otherwise. If

$$\psi(x) = \sum_n c_n \phi_n(x), \quad (3.14)$$

then multiplying by ϕ_m^* and integrating gives

$$\begin{aligned} \int \phi_m^*(x) \psi(x) dx &= \sum_n c_n \int \phi_m^*(x) \phi_n(x) dx \\ &= \sum_n c_n \delta_{mn} = c_m. \end{aligned} \quad (3.15)$$

Therefore,

$$c_m = \int_{-\infty}^{\infty} \phi_m^*(x) \psi(x) dx. \quad (3.16)$$

Orthogonality isolates one component of the superposition.

Questions.

Question 4. If $\psi(x) = \sum_n c_n u_n(x)$ is an expansion in specified orthonormal energy eigenfunctions $u_n(x)$, does the set of coefficients contain more information, less information, or the same information as $\psi(x)$?

- A) More information.
- B) Less information.
- C) The same information.
- D) It depends.

Answer and explanation. The answer is **C**. Once the basis is specified,

$$c_n = \int u_n^*(x) \psi(x) dx, \quad \psi(x) = \sum_n c_n u_n(x). \quad (3.17)$$

The wave function determines the coefficients, and the coefficients reconstruct the wave function.

3.16 Energy probabilities and interference

3.16.1 Interference between different energies

Consider two energy components,

$$\Psi(x, t) = \phi_1(x) e^{-i\omega_1 t} + \phi_2(x) e^{-i\omega_2 t}. \quad (3.18)$$

Its probability density is

$$\begin{aligned} |\Psi|^2 &= |\phi_1|^2 + |\phi_2|^2 \\ &\quad + \phi_1^* \phi_2 e^{-i(\omega_2 - \omega_1)t} + \phi_2^* \phi_1 e^{i(\omega_2 - \omega_1)t}. \end{aligned} \quad (3.19)$$

The last two terms are interference terms. They depend on the relative phase and are generally time-dependent.

Questions.

Question 5. For $\Psi_{\text{sum}} = \phi_1 e^{-i\omega_1 t} + \phi_2 e^{-i\omega_2 t}$, with $\omega_1 \neq \omega_2$, is the probability density time-independent?

- A) Yes, always.
- B) No, never.
- C) It depends on the wave functions and frequencies.

Answer and explanation. The answer is **C**. The density is generally time-dependent because the cross terms contain the changing phase $e^{\pm i(\omega_2 - \omega_1)t}$. At special positions, or for special wave functions, the cross terms can vanish.

3.16.2 Energy probabilities

If the energy eigenstates are orthonormal, the probability of measuring E_n is the squared modulus of its coefficient:

$$P(E_n) = |c_n|^2. \quad (3.20)$$

The time-dependent phase does not affect this probability because its modulus is one.

Questions.

Question 6. A state is

$$\Psi = \frac{i}{\sqrt{2}} e^{-i\omega_1 t} \psi_1 + \frac{1}{\sqrt{2}} e^{-i\omega_2 t} \psi_2,$$

where ψ_1 and ψ_2 are orthonormal energy eigenstates. What is the probability of measuring E_1 ?

- A) $\frac{1}{2} e^{-2i\omega_1 t}$
- B) $\frac{1}{\sqrt{2}}$
- C) $\frac{1}{2}$
- D) $\frac{1}{4}$

Answer and explanation. The answer is **C**:

$$P(E_1) = \left| \frac{i}{\sqrt{2}} e^{-i\omega_1 t} \right|^2 = \frac{1}{2}. \quad (3.21)$$

The factors i and $e^{-i\omega_1 t}$ are phases and have unit modulus.

3.17 Generalized eigenfunctions and continuous spectra

For a free particle,

$$\phi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}, \quad E(k) = \frac{\hbar^2 k^2}{2m}. \quad (3.22)$$

A plane wave is not square-normalizable:

$$\int_{-\infty}^{\infty} |\phi_k(x)|^2 dx = \infty. \quad (3.23)$$

Instead, plane waves satisfy delta normalization,

$$\int_{-\infty}^{\infty} \phi_k^*(x) \phi_{k'}(x) dx = \delta(k - k'), \quad (3.24)$$

and a normalizable wave packet is written as

$$\boxed{\psi(x) = \int_{-\infty}^{\infty} c(k) \phi_k(x) dk.} \quad (3.25)$$

The sum used for a discrete spectrum is replaced by an integral for a continuous spectrum.

3.18 The infinite square well

3.18.1 Physical setup and boundary conditions

A particle is confined between two perfectly rigid walls:

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x \leq 0 \text{ or } x \geq a. \end{cases} \quad (3.26)$$

The particle cannot be found outside the well, so

$$\phi(x) = 0 \quad \text{outside,} \quad \boxed{\phi(0) = \phi(a) = 0.} \quad (3.27)$$

These boundary conditions select the allowed energies.

3.18.2 Solve the stationary Schrödinger equation

Inside the well, $V = 0$, and

$$-\frac{\hbar^2}{2m} \frac{d^2\phi}{dx^2} = E\phi. \quad (3.28)$$

Defining

$$k^2 = \frac{2mE}{\hbar^2}, \quad (3.29)$$

gives

$$\frac{d^2\phi}{dx^2} + k^2\phi = 0. \quad (3.30)$$

The general solution is

$$\boxed{\phi(x) = A \sin(kx) + B \cos(kx).} \quad (3.31)$$

This is equivalent to $Ce^{ikx} + De^{-ikx}$, but the sine-cosine form makes the boundary conditions easier to apply.

At $x = 0$,

$$\phi(0) = B = 0, \quad (3.32)$$

so $\phi(x) = A \sin(kx)$. At $x = a$,

$$A \sin(ka) = 0. \quad (3.33)$$

The zero function is not a physical state, so $A \neq 0$. Therefore,

$$\sin(ka) = 0 \implies ka = n\pi, \quad n = 1, 2, 3, \dots \quad (3.34)$$

Thus

$$\boxed{k_n = \frac{n\pi}{a}, \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.} \quad (3.35)$$

Only an integer number of half-wavelengths can fit between the walls. The boundary conditions therefore convert the continuous variable k into discrete values.

3.18.3 Why there is no $n = 0$ state

For $n = 0$, the equation gives $\phi(x) = Cx + D$. The conditions $\phi(0) = \phi(a) = 0$ require $C = D = 0$, leaving only the zero function. Hence the lowest state is $n = 1$, with nonzero ground-state energy

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2} > 0. \quad (3.36)$$

3.18.4 Normalization and stationary states

For $0 < x < a$, write

$$\phi_n(x) = A_n \sin\left(\frac{n\pi x}{a}\right). \quad (3.37)$$

Normalization gives

$$\begin{aligned} 1 &= |A_n|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx \\ &= |A_n|^2 \frac{a}{2}, \end{aligned} \quad (3.38)$$

so $|A_n| = \sqrt{2/a}$. Choosing a real positive normalization constant,

$$\boxed{\phi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), & 0 < x < a, \\ 0, & \text{elsewhere.} \end{cases}} \quad (3.39)$$

The corresponding stationary state is

$$\psi_n(x, t) = \phi_n(x) e^{-iE_n t/\hbar}, \quad |\psi_n(x, t)|^2 = |\phi_n(x)|^2. \quad (3.40)$$

3.18.5 Physical interpretation

The allowed wavelengths are

$$\lambda_n = \frac{2a}{n}, \quad (3.41)$$

and the n -th state has $n - 1$ internal nodes. Higher-energy states oscillate more rapidly and have more nodes. The energy scale is

$$E_n \propto \frac{n^2}{ma^2}. \quad (3.42)$$

For the first three states,

$$E_1 : E_2 : E_3 = 1 : 4 : 9. \quad (3.43)$$

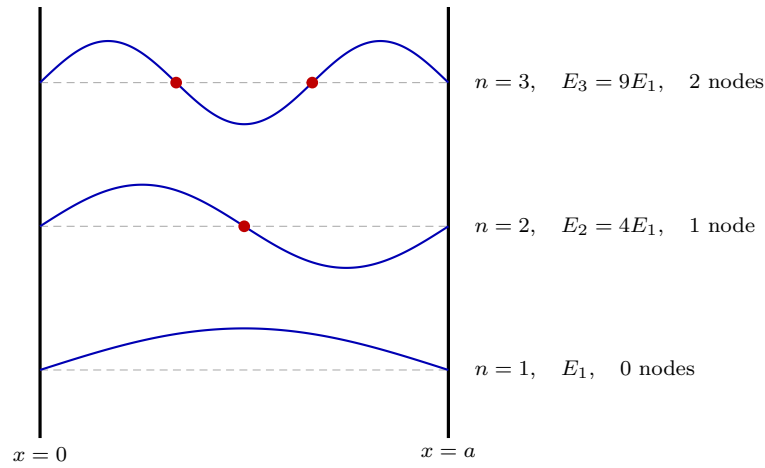


Figure 3.1: The first three eigenfunctions in one infinite square well. The curves are vertically offset. Red dots mark internal nodes; wall zeros are not counted.

The figure shows the general pattern:

$$\text{the } n\text{-th state has } n - 1 \text{ internal nodes, } E_n = n^2 E_1. \quad (3.44)$$

A narrower well produces larger energies because stronger localization requires shorter wavelengths and greater kinetic energy.

Inside the well,

$$\sin(k_n x) = \frac{e^{ik_n x} - e^{-ik_n x}}{2i}. \quad (3.45)$$

The eigenfunction is a standing wave containing components moving in both directions, not a single plane wave with one definite momentum direction.

Questions.

Question 7. How does the ground-state energy of an infinite square well compare with that of a wider well?

- A) The wider well has lower energy.
- B) The wider well has higher energy.

- C) The two energies are the same.
- D) More information is needed.

Answer and explanation. The answer is **A**. Since

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2} \propto \frac{1}{a^2}, \quad (3.46)$$

increasing the width lowers the characteristic momentum and ground-state energy.

3.19 Bound-state structure

The infinite square well illustrates general features of bound states:

$$\boxed{\text{boundary conditions and normalizability} \implies \text{discrete energies.}} \quad (3.47)$$

For a smooth one-dimensional potential,

$$\frac{\phi_E''}{\phi_E} = -\frac{2m}{\hbar^2}(E - V). \quad (3.48)$$

Therefore,

$$E > V : \text{oscillatory behavior}, \quad E < V : \text{exponential behavior.} \quad (3.49)$$

In one-dimensional bound-state problems, the ground state has no internal node, and higher-energy states have more nodes and greater curvature.

Lecture 9 summary

- One energy eigenstate is stationary; a superposition of different energies generally has time-dependent interference.
- For an orthonormal basis,

$$c_n = \int \phi_n^*(x) \psi(x) \, dx, \quad P(E_n) = |c_n|^2.$$

- Plane waves are delta-normalized generalized eigenfunctions; normalizable free-particle states are wave packets.
- Boundary conditions in the infinite square well give

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.$$

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., sections on stationary states, expansion in energy eigenfunctions, and the infinite square well. Review Lecture 8 on unitary evolution and the general energy-basis solution.